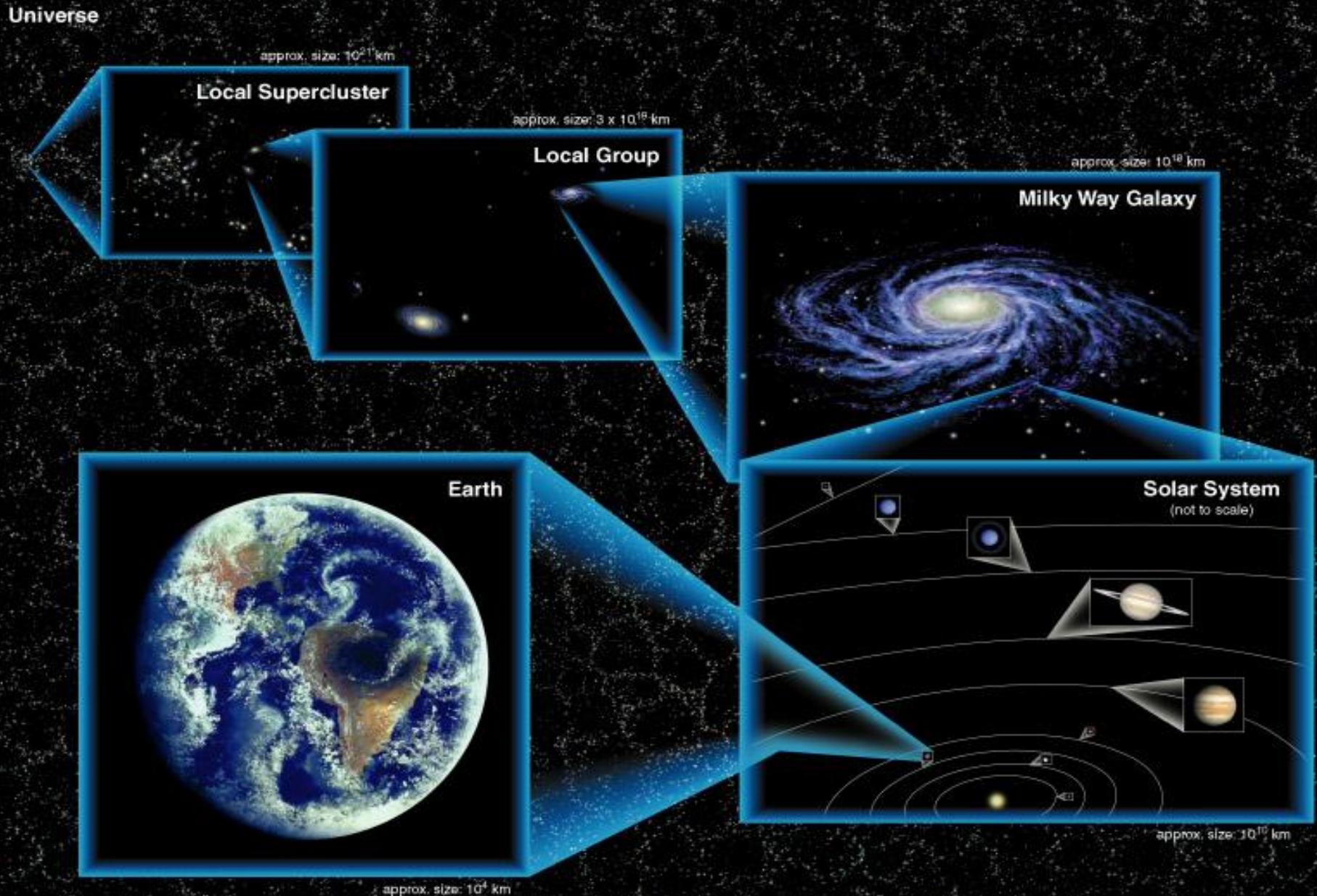




# Motion of the Sky

# Lecture 1 Review (Where are we?)



# The Night Sky

Goals for Today:

- 1) What does the Universe look like from Earth?
- 2) What are constellations?
- 3) What is the Celestial Sphere?
- 4) How does the sky move (daily)?

Words to know: Asterism, celestial north and south poles, horizon, zenith, altitude, circumpolar

## What does the Universe look like from Earth?

With the naked eye, we can see more than 2,000 stars. We can also see the Milky Way, most of the planets, many comets, a whole bunch of satellites, and some galaxies.



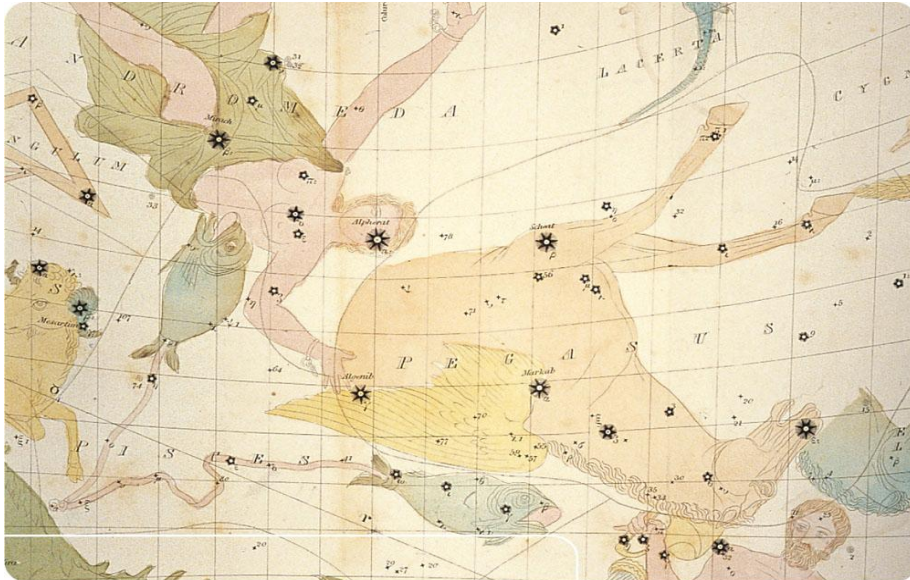
# Constellations – Modern Definition

A **constellation** is a loose grouping of stars that appear close together in the night sky

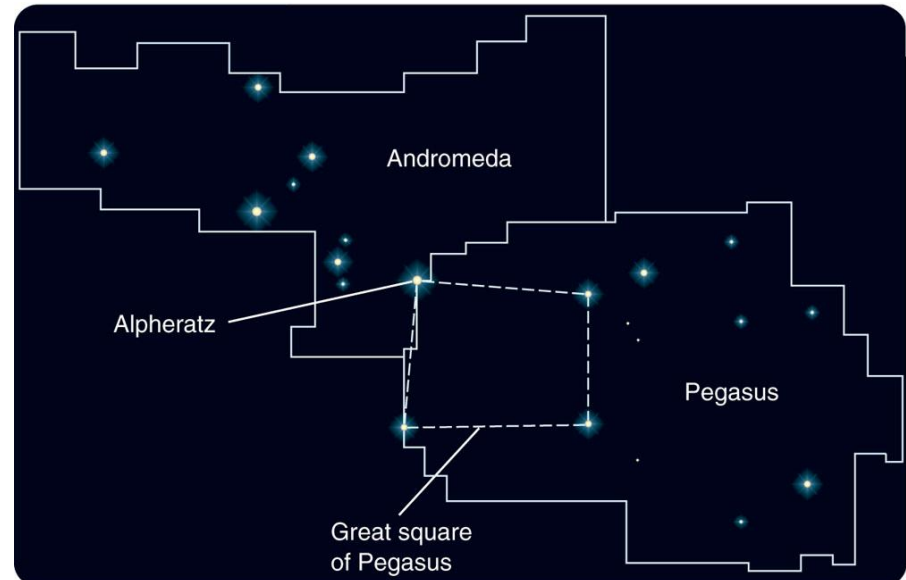
In ancient times, they included only bright stars.

Today, there are an agreed-upon set of 88 constellations that together divide the entire sky

- Like a country boundary, “the star's address”



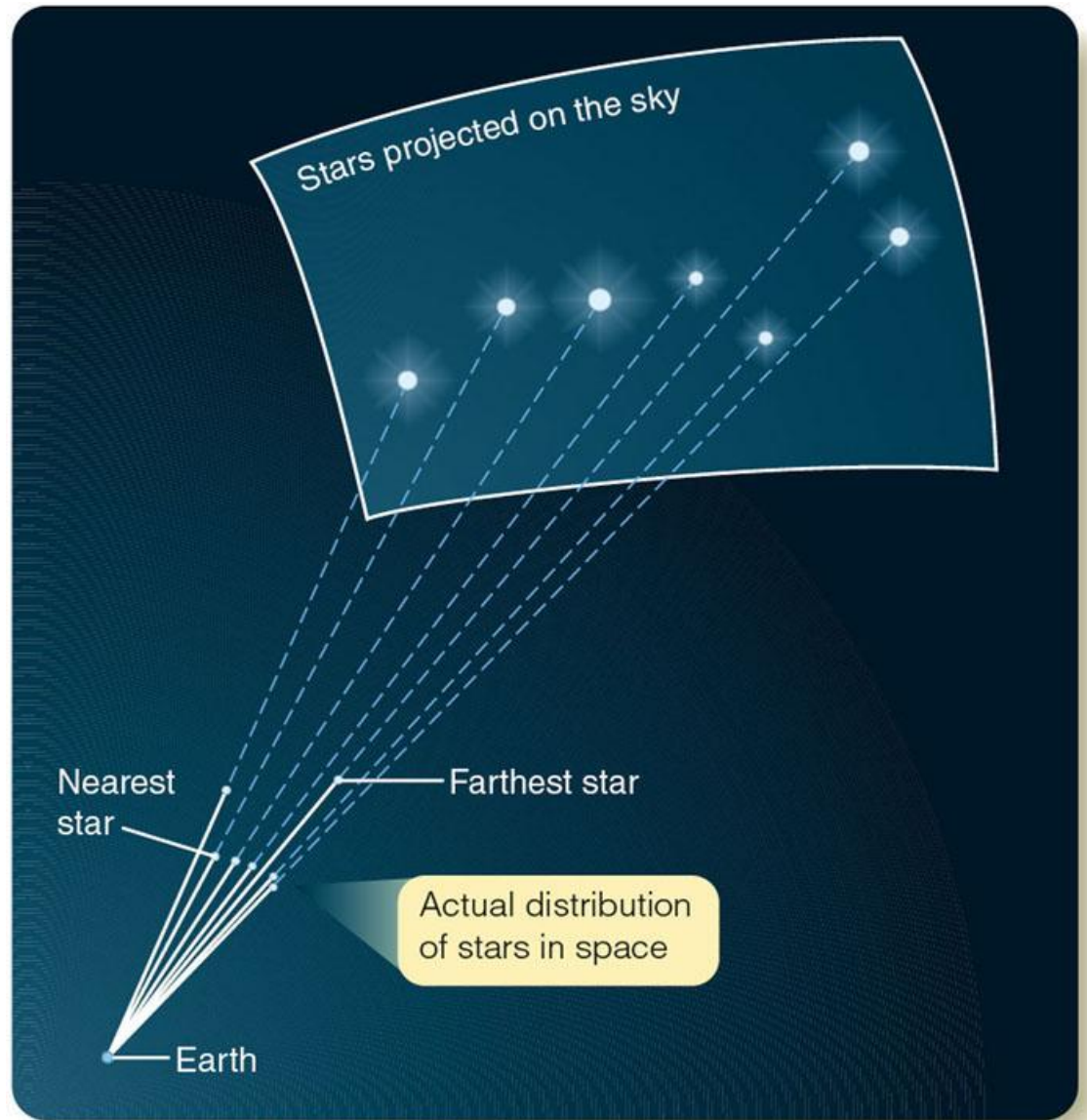
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# Constellations Are Not Real Groups

The stars in constellations and asterisms are not necessarily physically close to each other.



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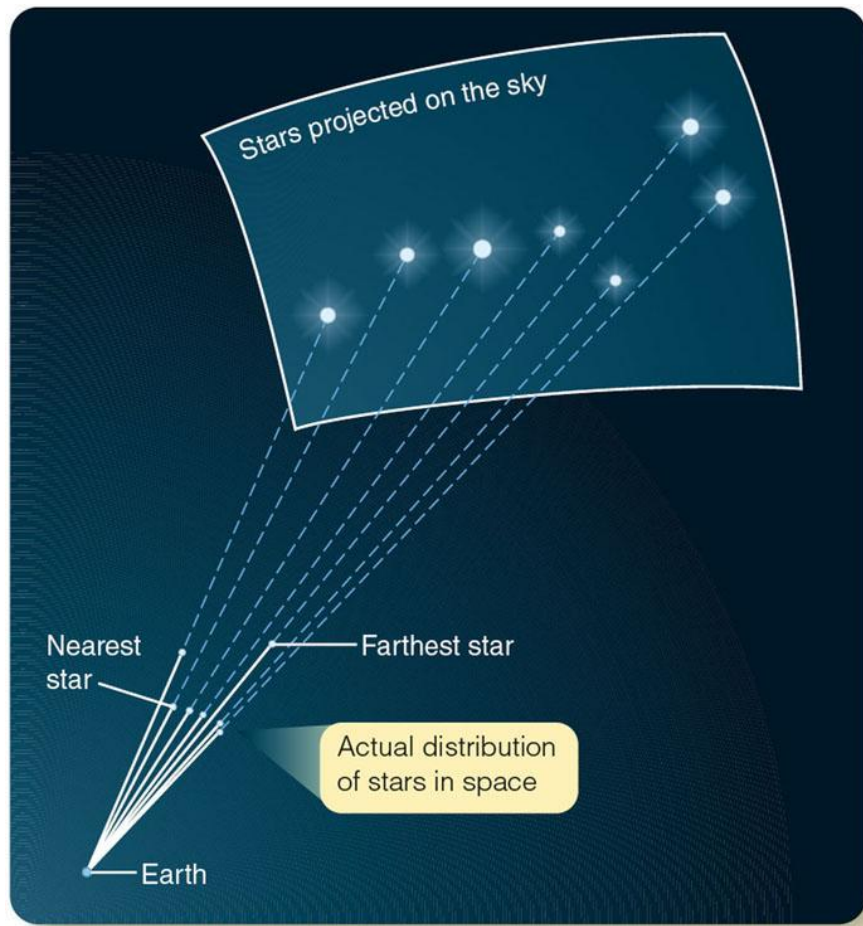
# The Celestial Sphere



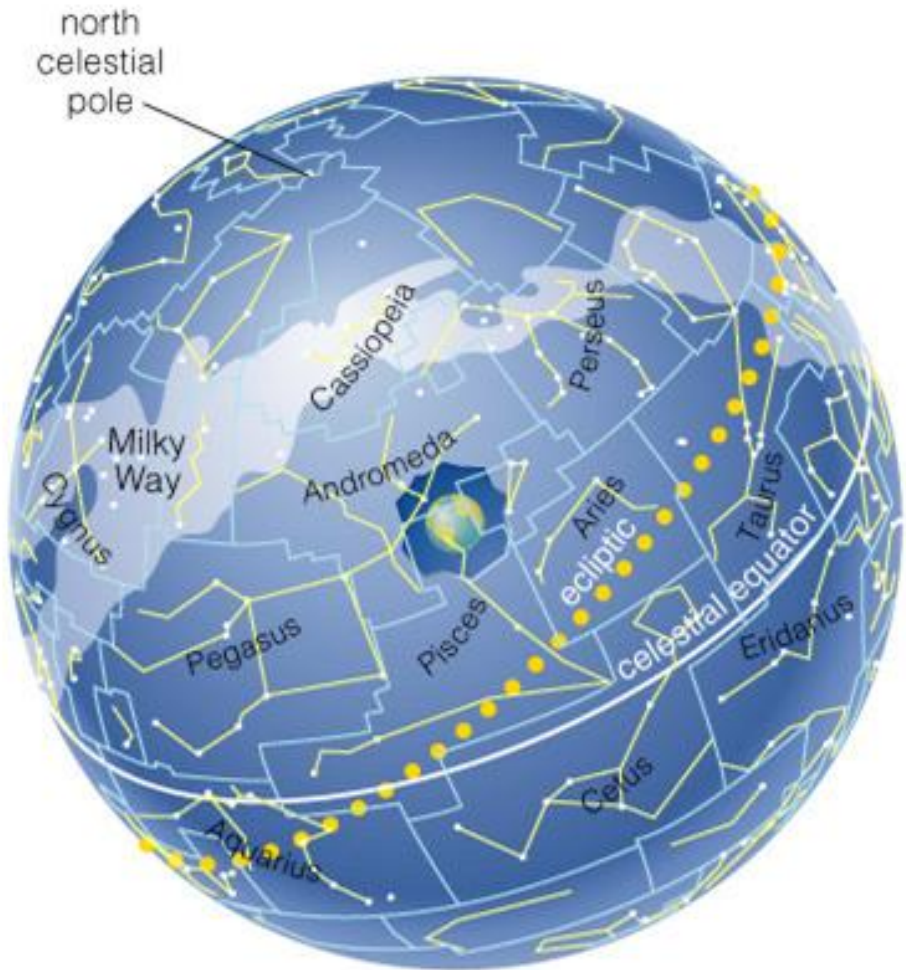
The **celestial sphere** is a *model* of the sky that helps us visualize the location and motion of stars.

# The Celestial Sphere is not Real!

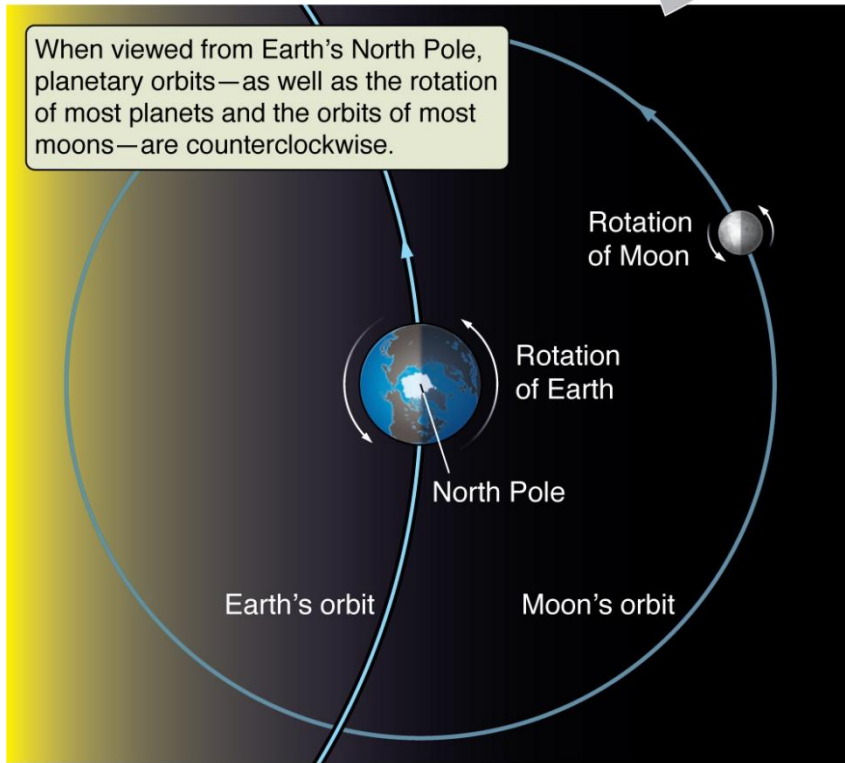
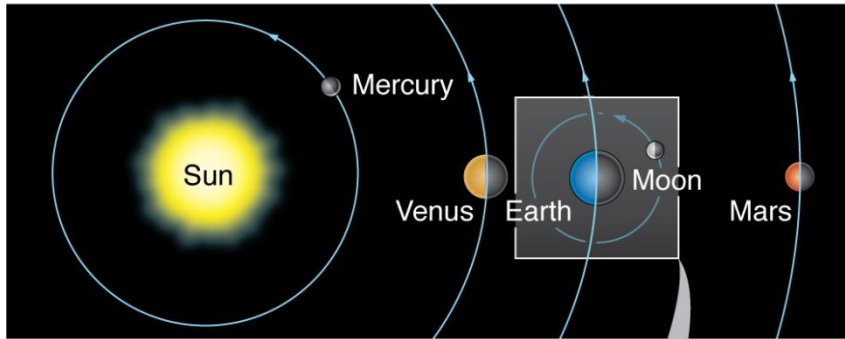
It is a model - the stars are not literally on a sphere.



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# Earth Rotates on its Axis



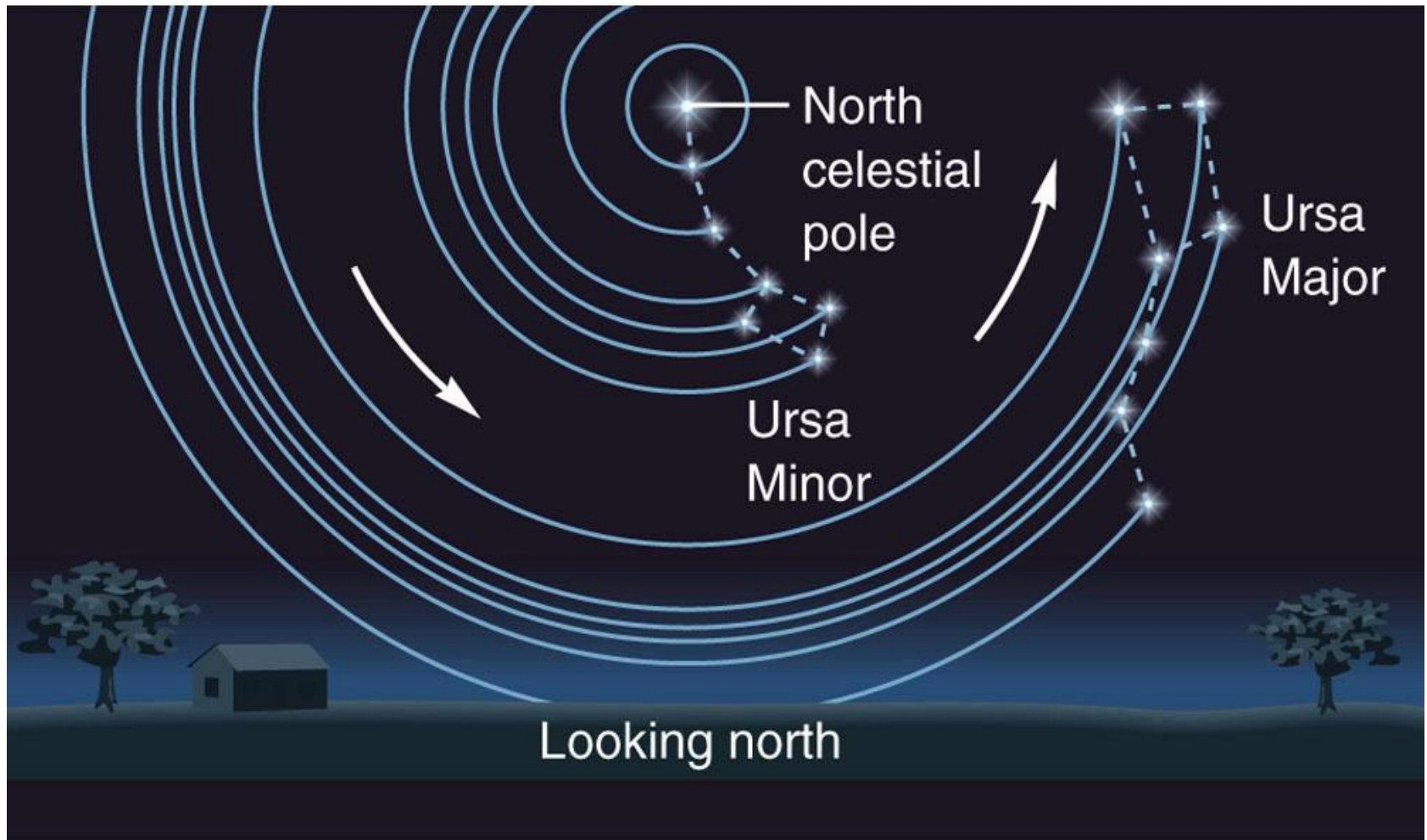
There are two main motions that affect what we see in the sky:

## Motion #1:

The Earth rotates on its axis. This causes stars to appear to rotate around the north and south *celestial* poles. The complete rotation of each star takes 24 hours.

(Motion #2 is due to the revolution of the Earth)

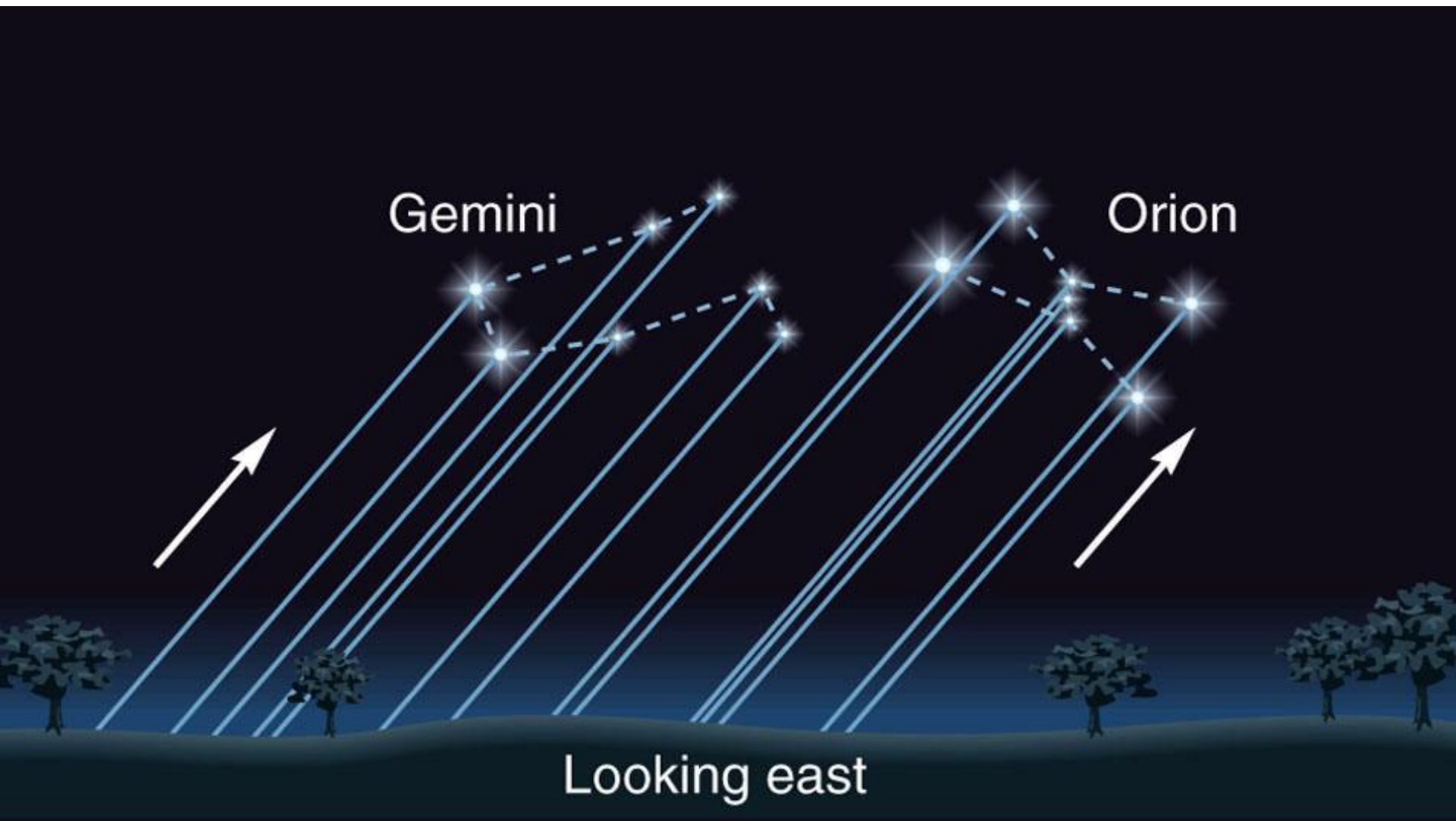
## Looking North (from the Northern Hemisphere)



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Some of these stars are **circumpolar**, never rising or setting.

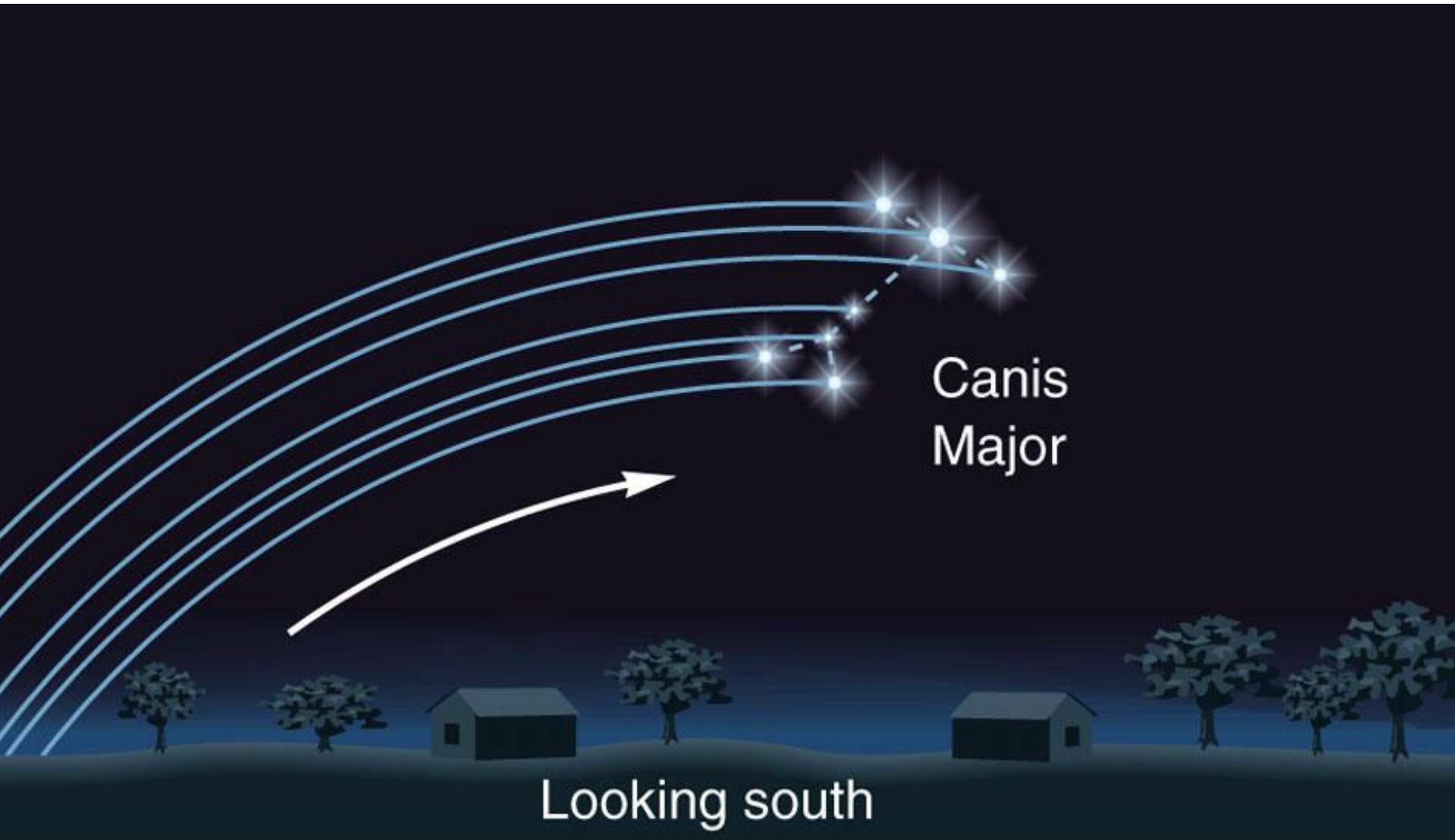
## Looking East (from the Northern Hemisphere)



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## Stars rise in the East....

## Looking South (from the Northern Hemisphere)



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Pass through the South and set in the West.

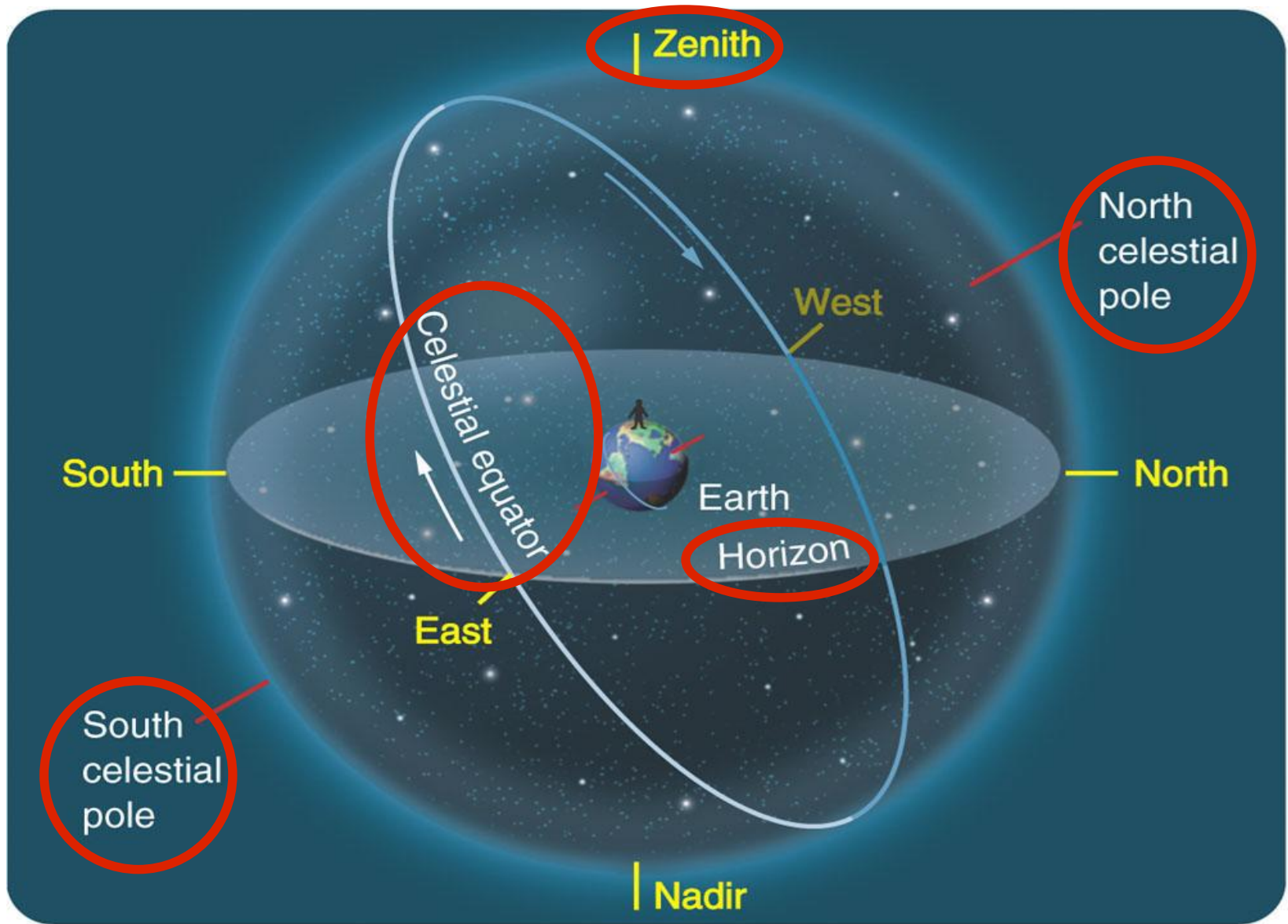
# VIDEOS!!!!

<http://www.youtube.com/watch?v=3V3rmDG5J8A>

<http://www.youtube.com/watch?v=uHcRPJFR47Q>

<http://www.youtube.com/watch?v=ANAmiuAkyGQ>

# Celestial Sphere



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# Object Rise/Set Directions

1) The direction an object rises determines where it sets

An object rising due East will set due West

An object rising South-East will set in the South-West

An object rising North-East will set North-West

2) The direction an object rises determines the amount of time it is visible.

# Concept Quiz

Which direction is the observer facing in this picture (photo was taken from the northern hemisphere)?

- a) North
- b) East
- c) South
- d) West



# Concept Quiz

Which direction is the observer facing in this picture (photo was taken from the northern hemisphere)?

- a) North
- b) East
- c) South
- d) West



# In your Lecture-Tutorials Book

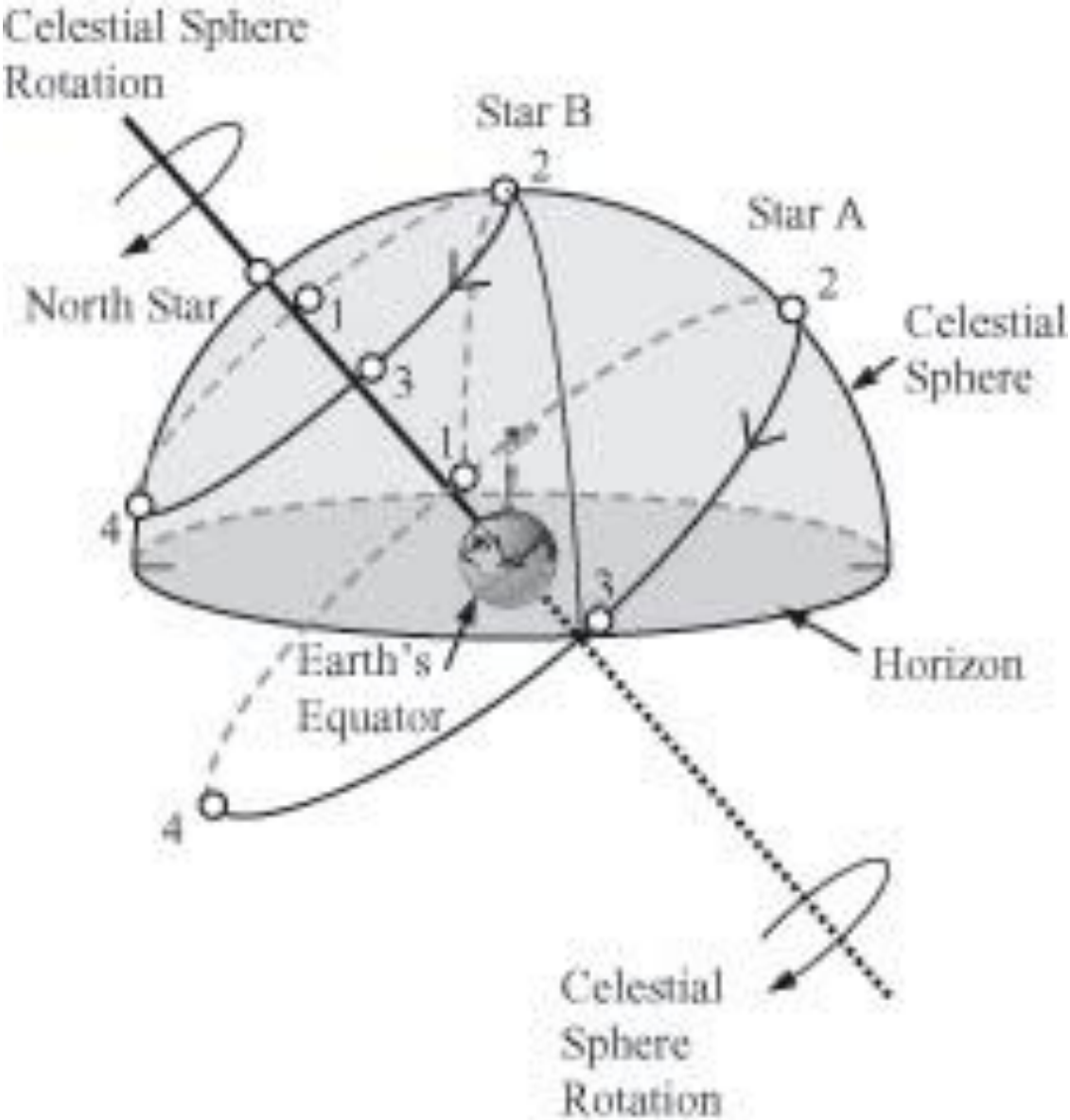
Do “Position”, page 1, #1-9, 11 (skip #10)

Work in pairs or small groups

Ask other groups for help

Have fun!

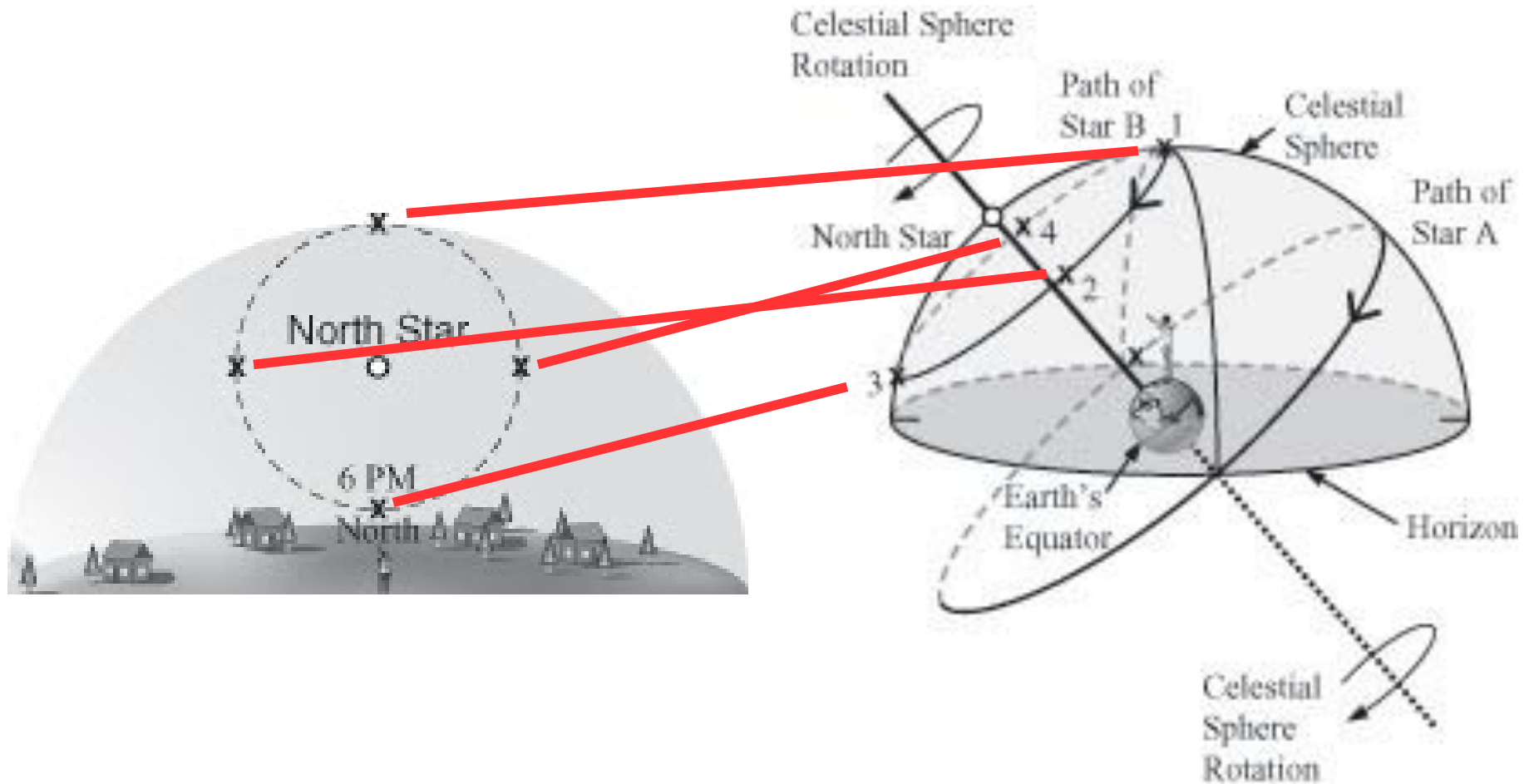
I'll give you ~10 min.



# More!

## Lecture-Tutorials “Motion” #1-6

# LT - Motion



# Clicker Quiz

Please put these rising stars in order from least time above horizon to most time (A is circumpolar)

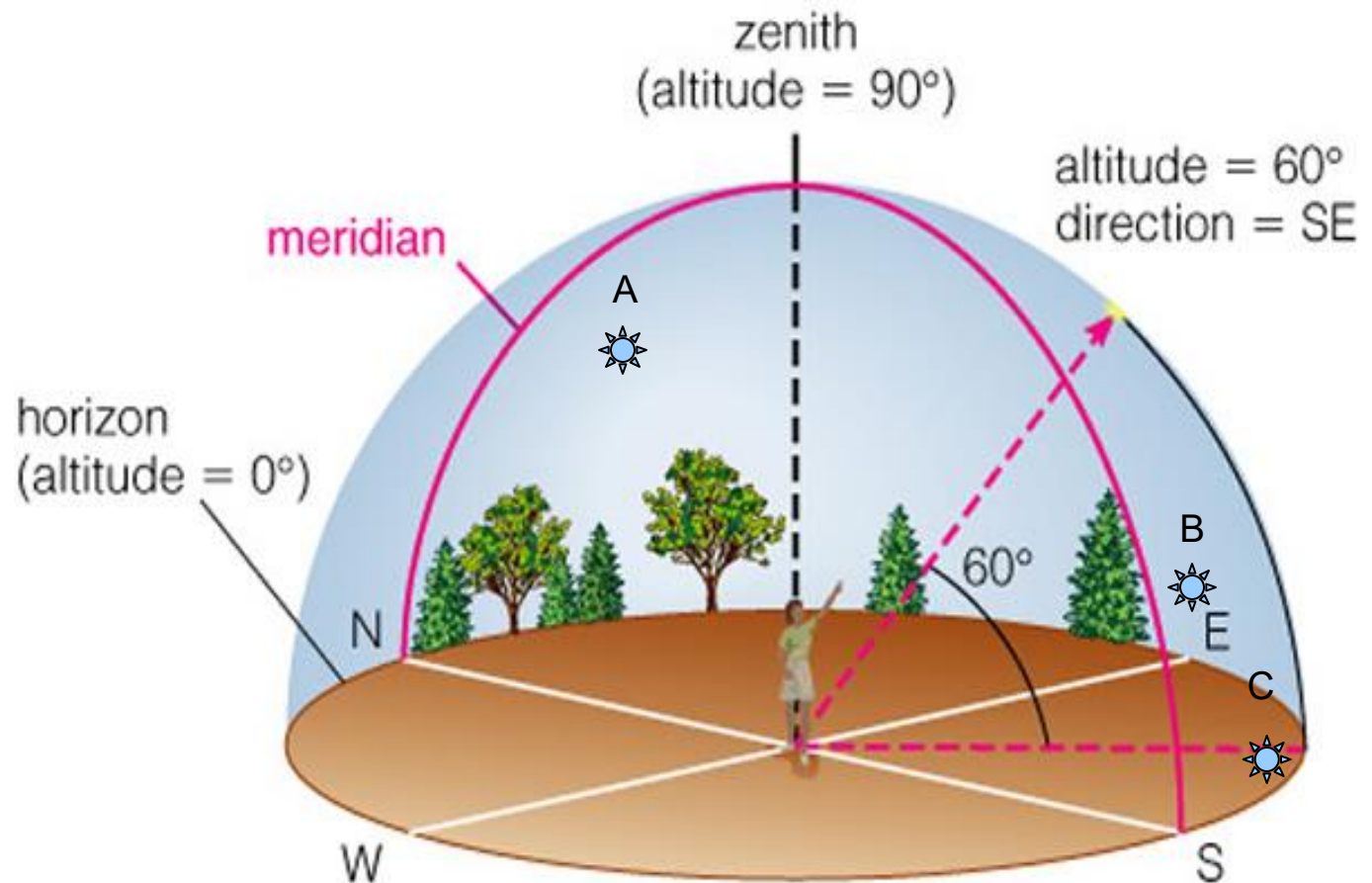
a) A, B, C

b) C, A, B

c) B, A, C

d) C, B, A

e) A, C, B



# Clicker Quiz

Please put these rising stars in order from least time above horizon to most time (A is circumpolar)

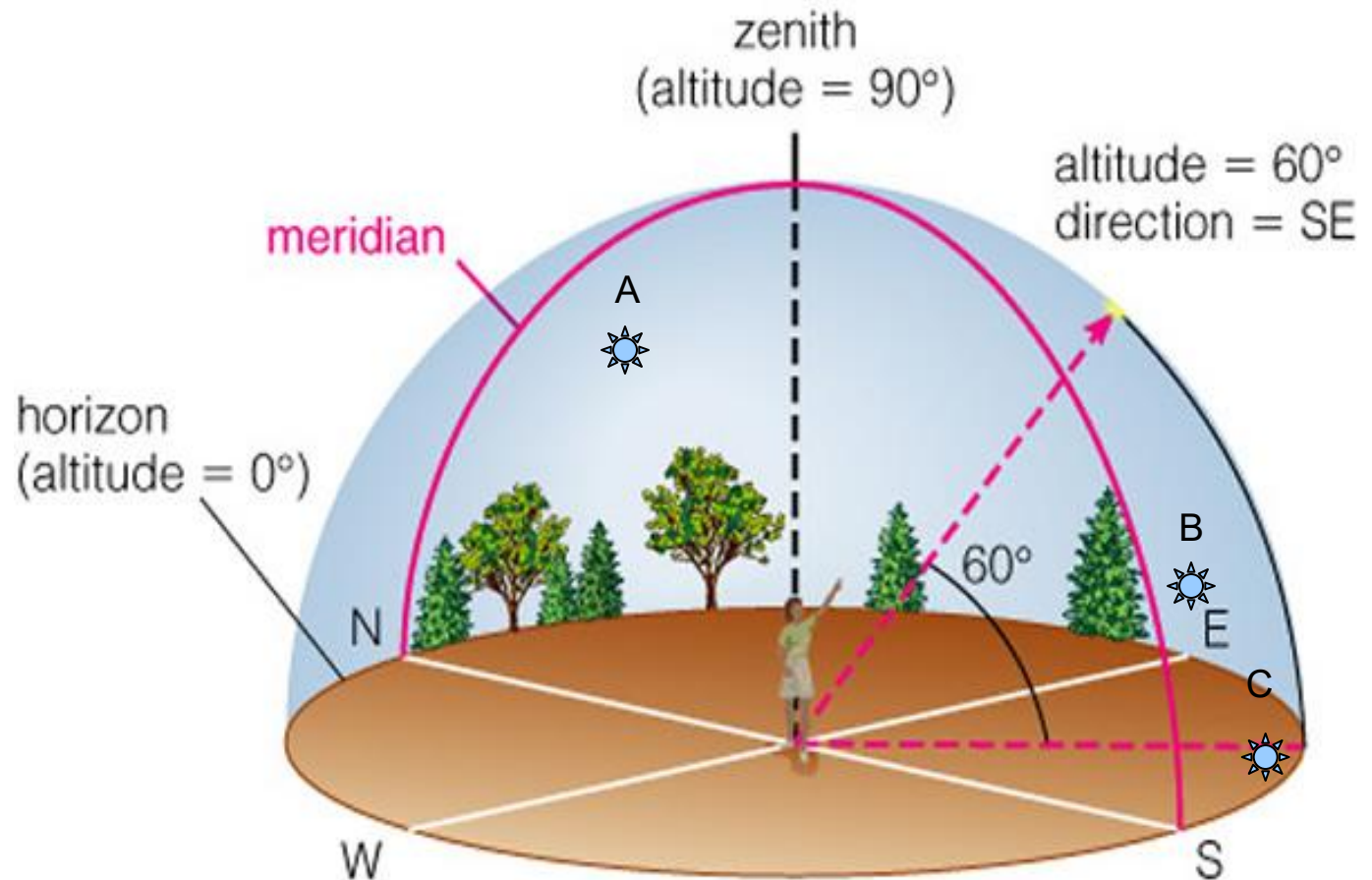
a) A, B, C

b) C, A, B

c) B, A, C

d) C, B, A

e) A, C, B



# Lecture 2 Review – Time above Horizon

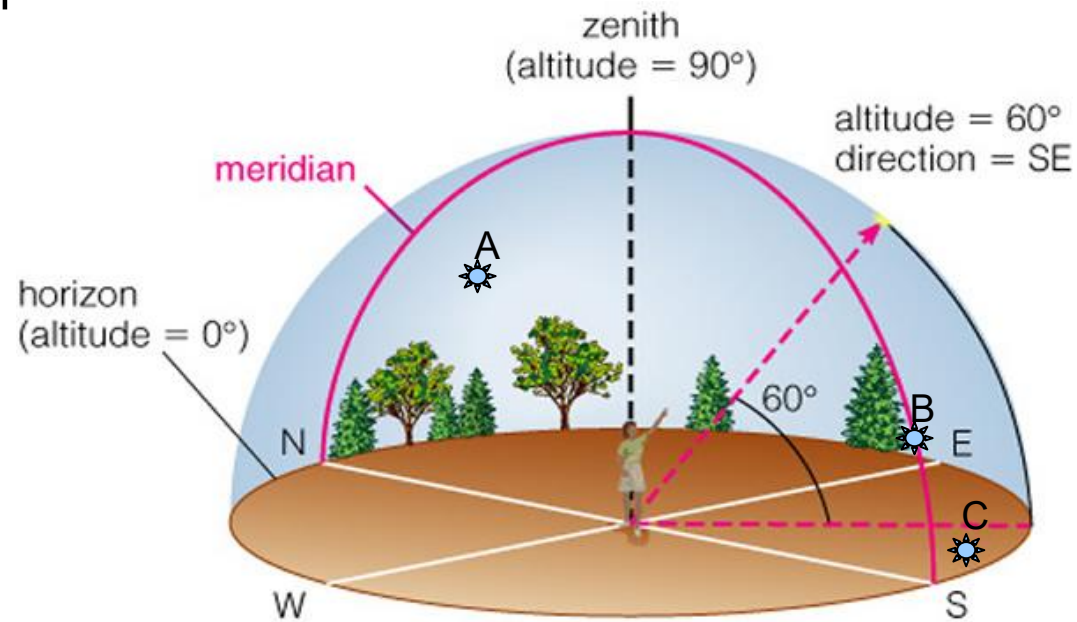
Every star takes 24 hours to complete a circle.

Each star is above the horizon for a different amount of time.

A for 24 hours

B for about 12 hours

C for less than 12 hours



# Celestial Sphere Definitions

**Celestial equator:** the projection of the Earth's equator onto the sphere.

**Horizon:** the limit to what can be seen from a particular vantage point

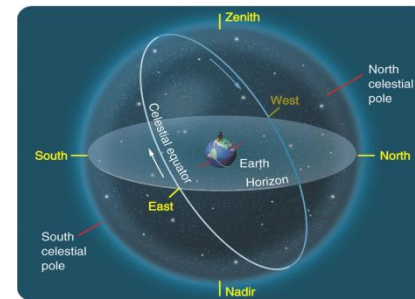
**North/South Celestial Poles:** the projection of the North/South Poles onto the Celestial Sphere

**Zenith:** the point directly overhead

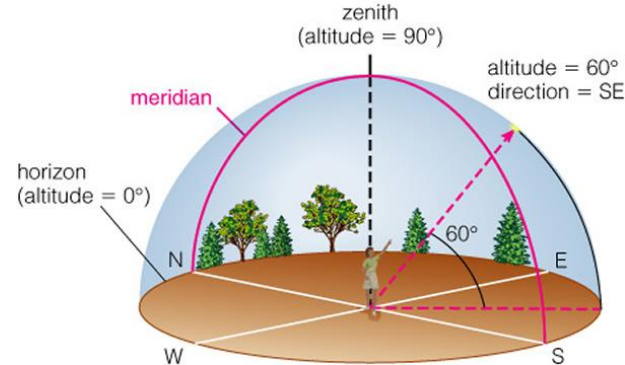
# Three Views

You should be able to go back and forth between three different views of the sky:

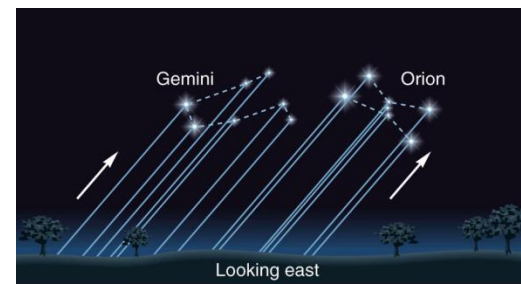
1) Above Earth



1) Above the viewer



1) Viewer's perspective



# The Celestial Sphere Moves! (Daily)

Main points:

- 1) The motion of the stars can be understood as the 24 hour motion of the celestial sphere.
- 2) We see stars (and the Moon and the Sun) rise in the East and set in the West <- Important!
- 3) Stars appear to orbit the north celestial pole. In the north, there is a star very near the north celestial pole called the “North Star” or Polaris
- 4) Some stars are always visible - these are called “circumpolar”
- 5) Some stars are never visible!

What we see depends on the direction we are facing!

# Animations

[https://astro.unl.edu/naap/motion2/animations/ce\\_hc.html](https://astro.unl.edu/naap/motion2/animations/ce_hc.html)

Things to notice:

- 1) Circumpolar, and “not visible” regions
- 2) How these change with latitude on Earth

<https://www.solarsystemscope.com>

Things to notice:

- 1) Sun’s position with respect to the signs of the zodiac and how this changes throughout the year
- 2) Sun’s location with respect to the signs of the zodiac does not change throughout one day

# The Night Sky

Goals for Today:

- 1) What does the Universe look like from Earth?
  - We can see ~2000 stars, a few Galaxies and the Milky Way
- 2) What are constellations and asterisms?
  - Constellations are regions on the sky enclosing groupings of stars. Asterisms are less-formal groupings of stars
- 3) What is the Celestial Sphere?
  - The projection of all of space onto a sphere that we are inside of

# A User's Guide to the Night Sky

## 4) How do stars move?

- Stars rise in the East and set in the West.
- They appear to circle around the North Celestial Pole (in the northern hemisphere)
- Their entire motion takes 24 hours
- The location where the rise determines where they will set
- Some stars are never visible

Words to know: Asterism, celestial north and south poles, horizon, zenith, altitude, circumpolar

# More motions of the sky

## Goals for Today:

- 1) How do the star's motions change with latitude?
- 2) What is the significance of the signs of the Zodiac and the ecliptic
- 3) Why do the constellations change throughout the year?

## Concept Quiz

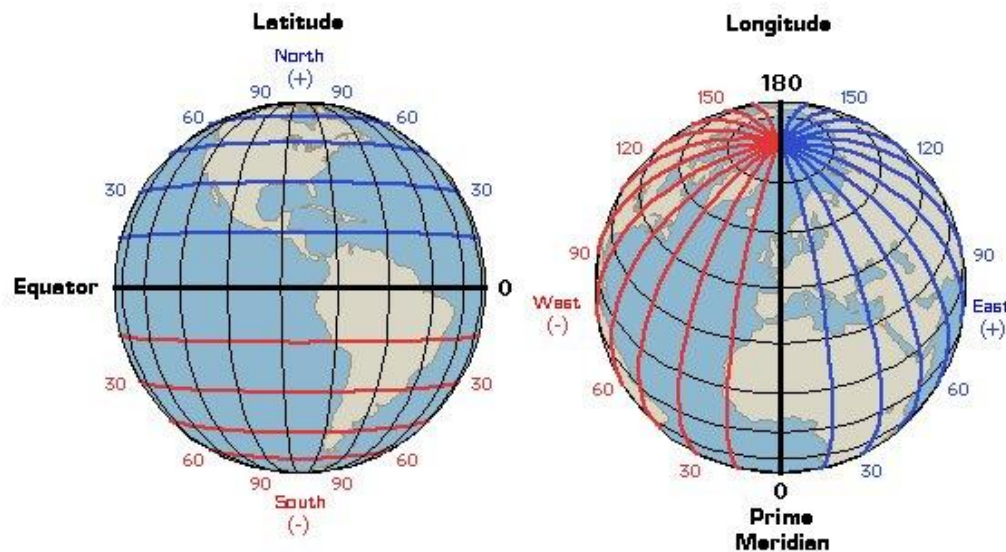
What is latitude? A coordinate that...

- a) ...specifies position in the north-south direction
- b) ...specifies position in the east-west direction
- c) ...specifies position in the vertical direction

# Concept Quiz

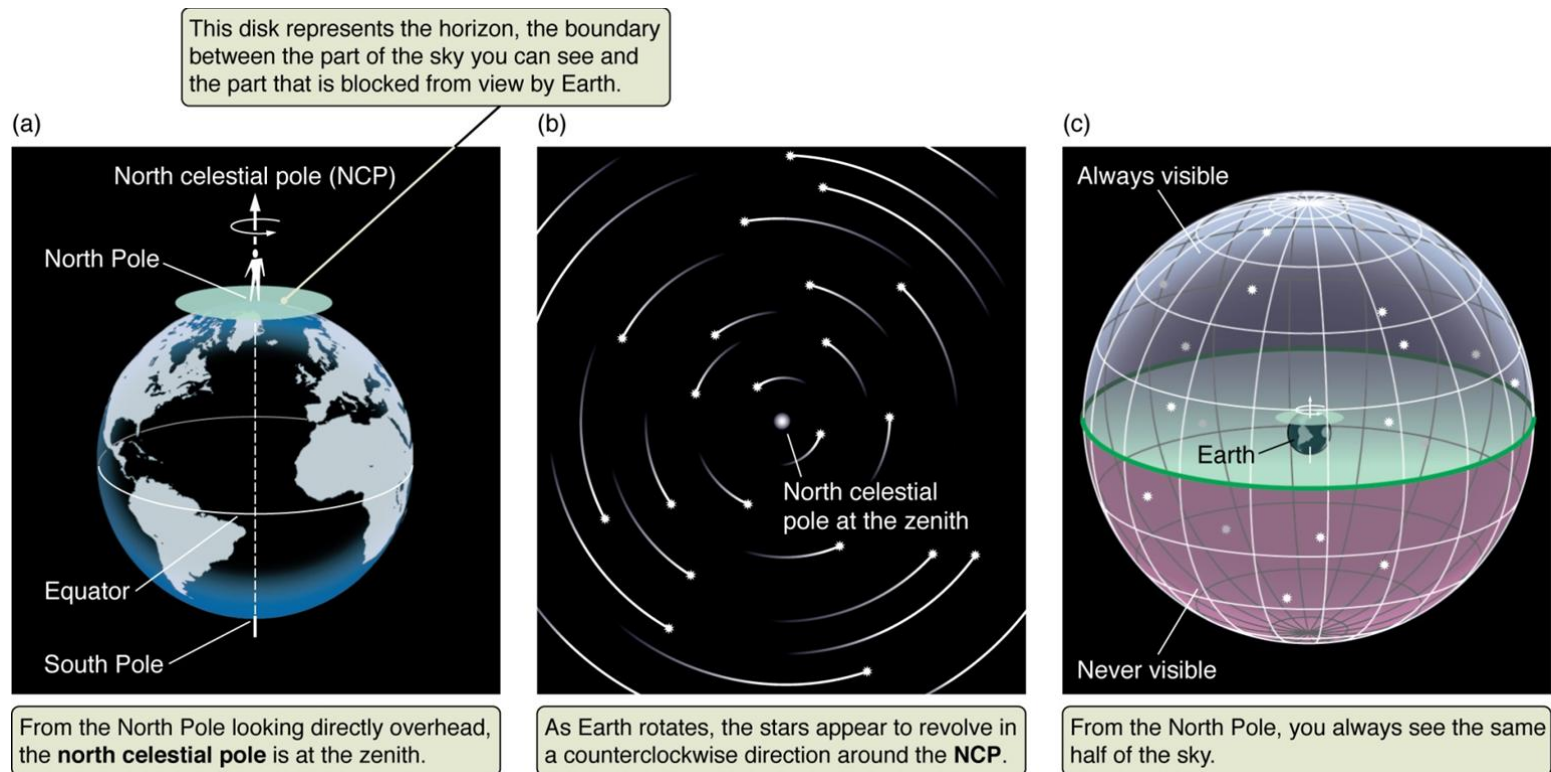
What is latitude? A coordinate that...

- a) ...specifies position in the north-south direction
- b) ...specifies position in the east-west direction
- c) ...specifies position in the vertical direction



# Motion of Stars at the North Pole

The north celestial pole is directly overhead.  
 Stars rotate counterclockwise in about 24 hours.  
 No stars rise or set: all are *circumpolar*.



# Motion of the Sky at the North Pole

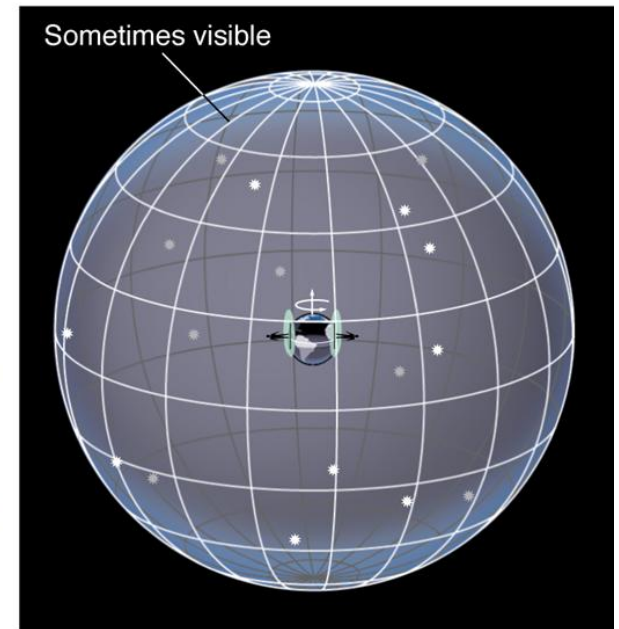
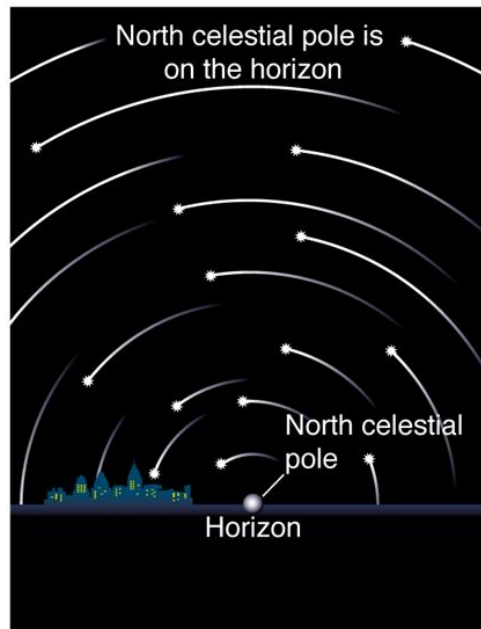
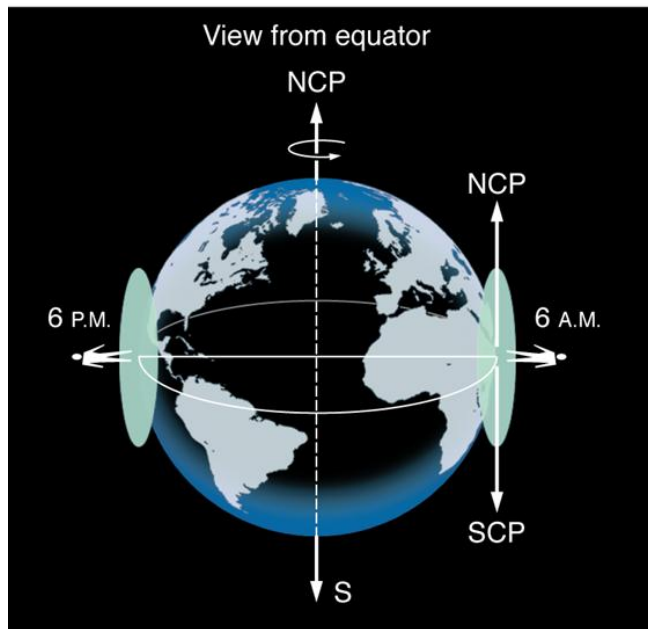


# Motion of Stars at the Equator

All stars rise and set.

The celestial poles are on the northern and southern horizons.

Observers can see the whole celestial sphere as it rotates.



# Motion of Stars at the Equator



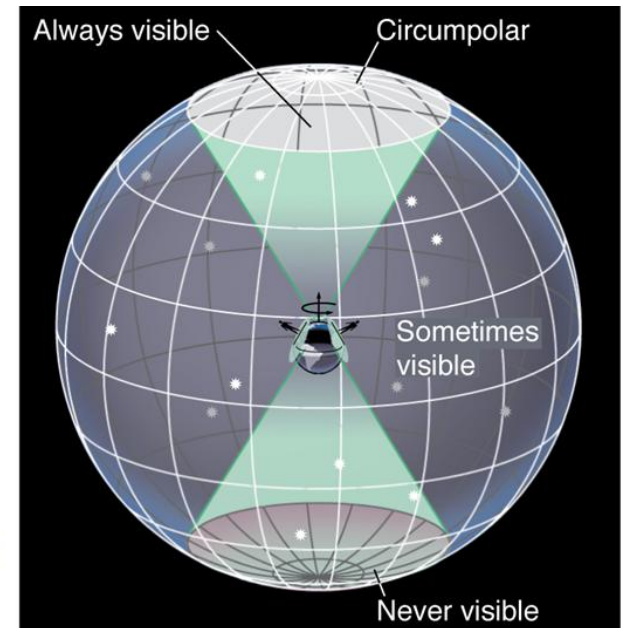
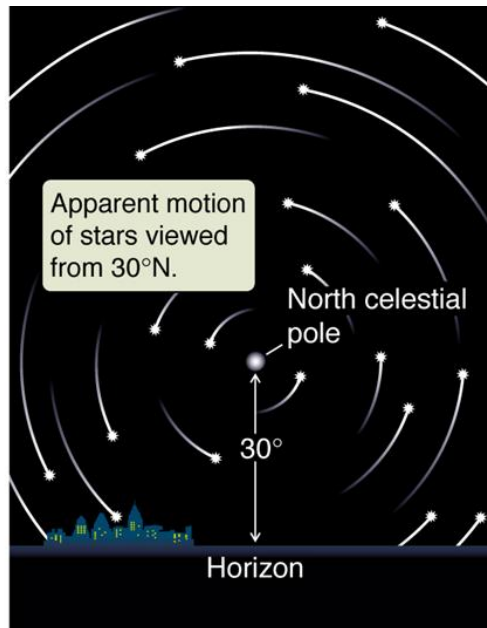
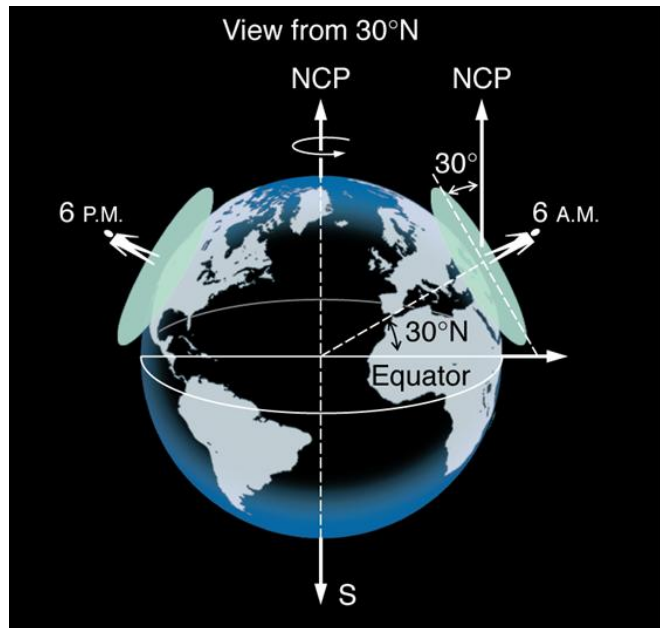
# Motion of Stars Between Poles and Equator (Morgantown)

Some stars are circumpolar.

Some stars rise and set.

Some stars are never visible.

**THE ALTITUDE OF THE POLE STAR IS YOUR LATITUDE!**



# Revolution versus Rotation

Motion #1: The daily motion of the stars in the sky is caused by the rotation of the Earth on its axis.

- This causes the daily motion of the star across the sky.

Motion #2: The yearly motion of the Earth around the Sun is called revolution.

- This causes the change in which stars are visible at night throughout the year.
- > Revolvers rotate!

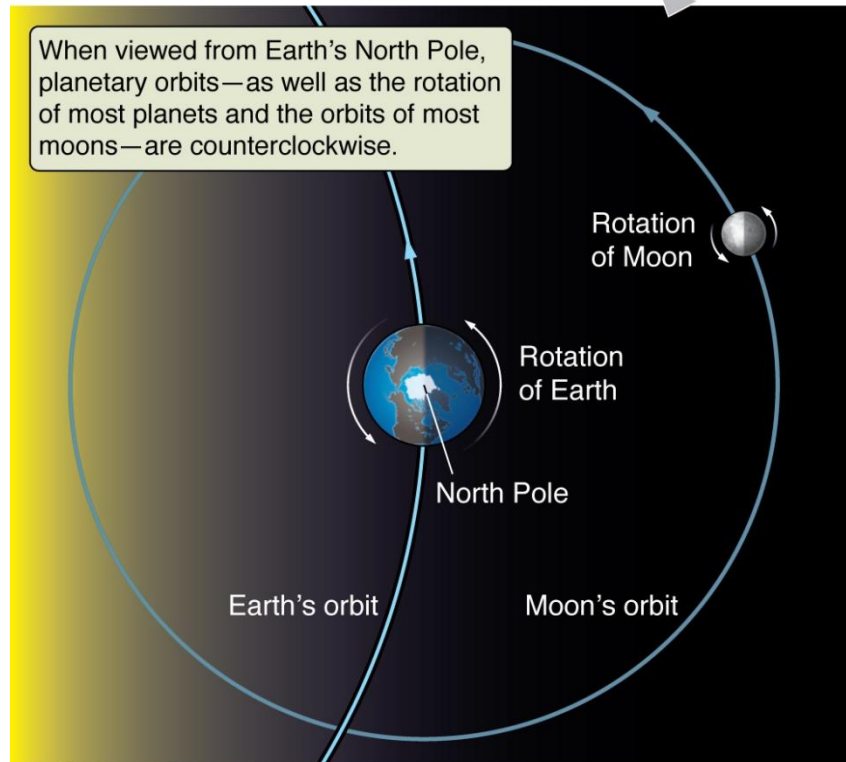
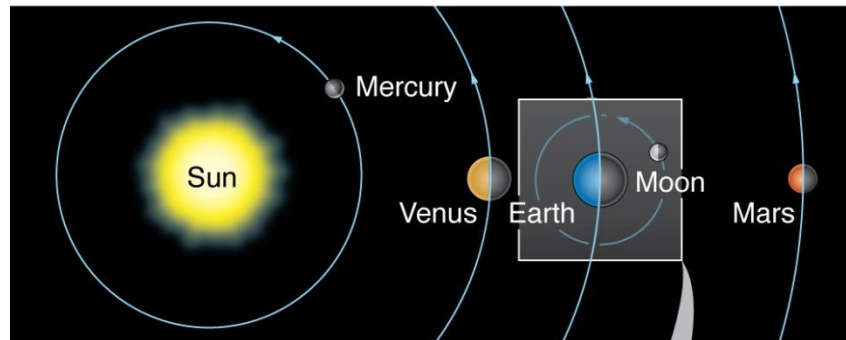
# Earth Revolves around the Sun

## Motion #2 – Earth revolving around the Sun (yearly)

Takes 365.242 days

We have a “leap year” every four years

- 365.25 days on average
- Unless if year divisible by 100
  - 365.240
  - Unless divisible by 400
    - 365.242

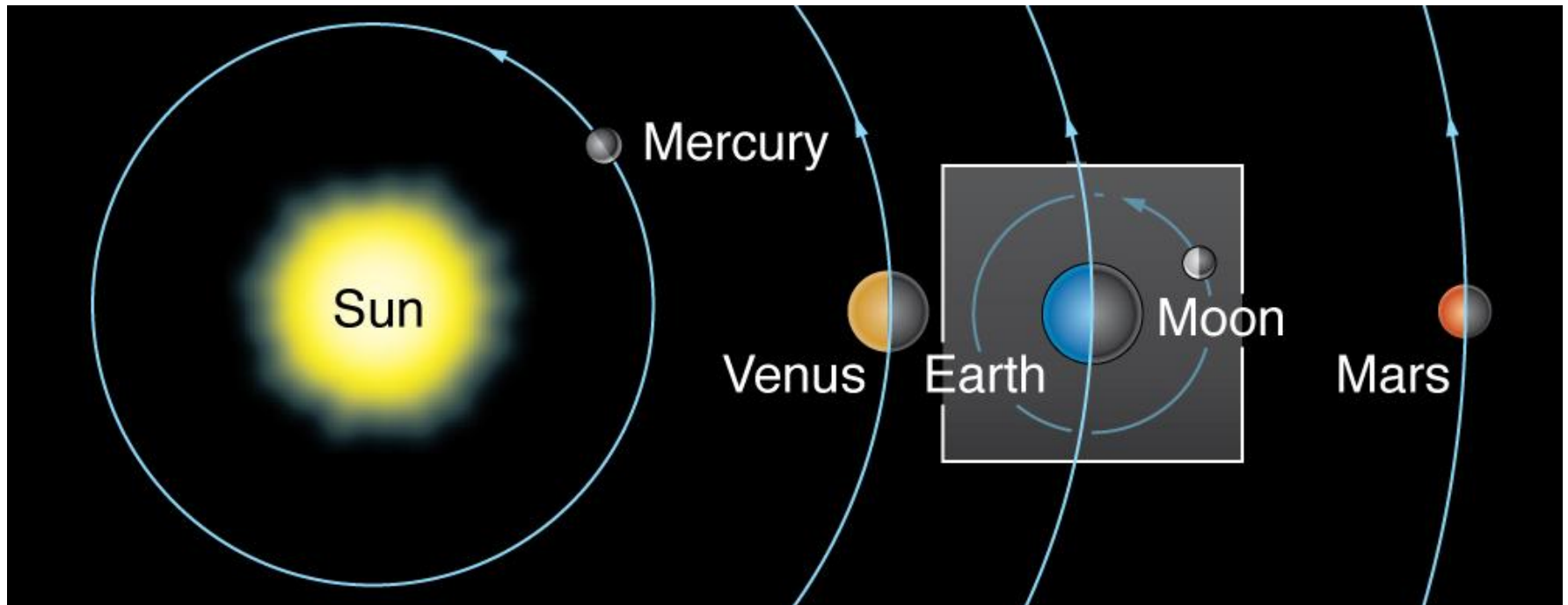


# The Size of Earth's Orbit

Earth's orbit is nearly circular (within a couple percent).

The average distance to the Sun is called the astronomical unit, or AU. <- Vocab word of the day!

- 1 AU = 150 million km = 93 million miles (million =  $10^6$ )



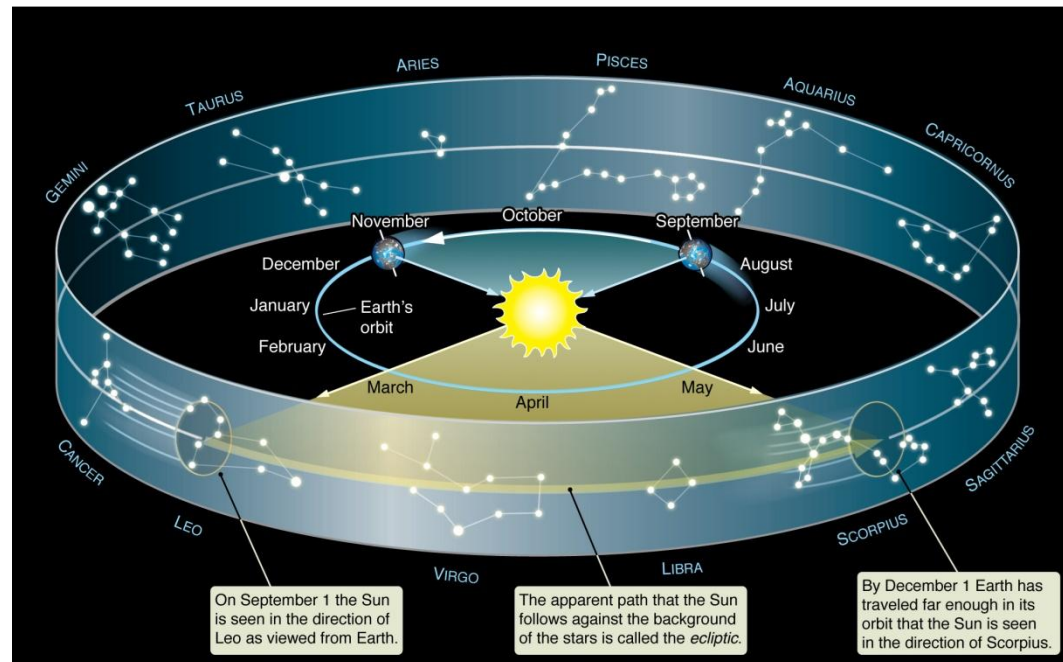
# The Zodiac

As Earth revolves about the Sun, the Sun is seen against different constellations

These constellations are called the signs of the zodiac.

The constellations (and stars) we see depend on the season (motion #2)

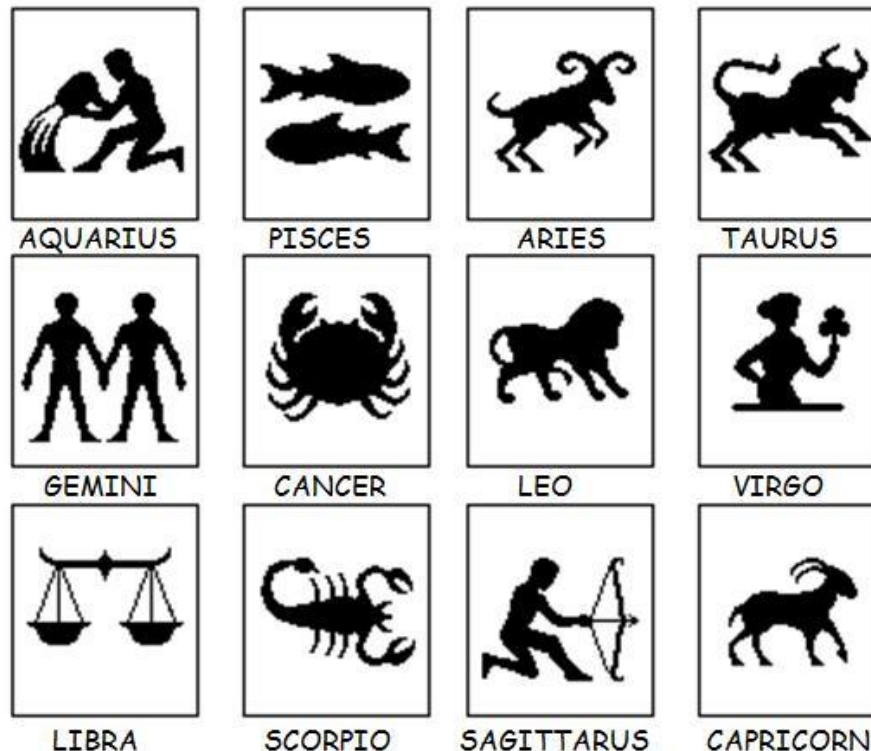
- “Winter” constellations are up during the day in summer



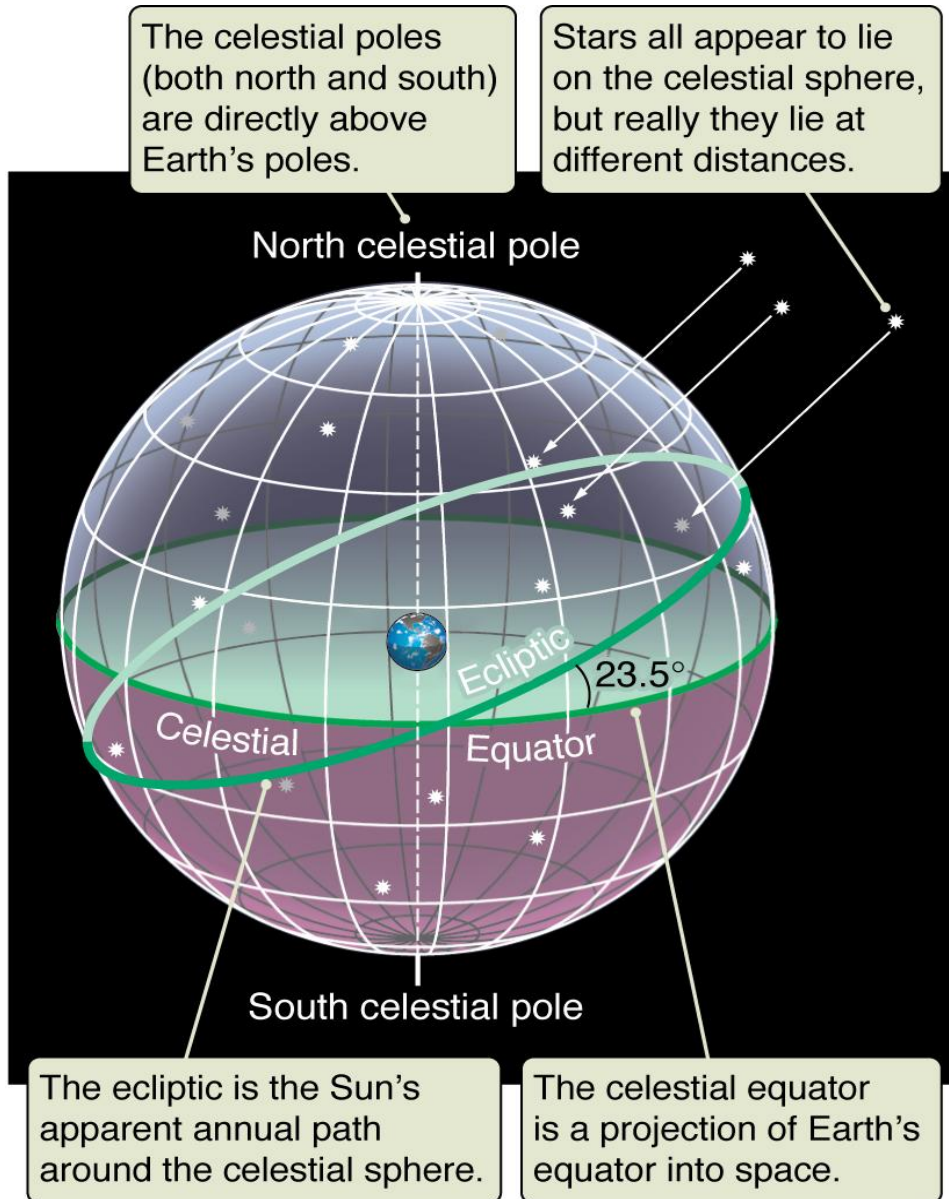
# Signs of the Zodiac

The Babilonians first identified constellations along the ecliptic – the signs of the zodiac

**Important: astrology, which uses the signs of the zodiac, is not science! Do not confuse it with astronomy!**



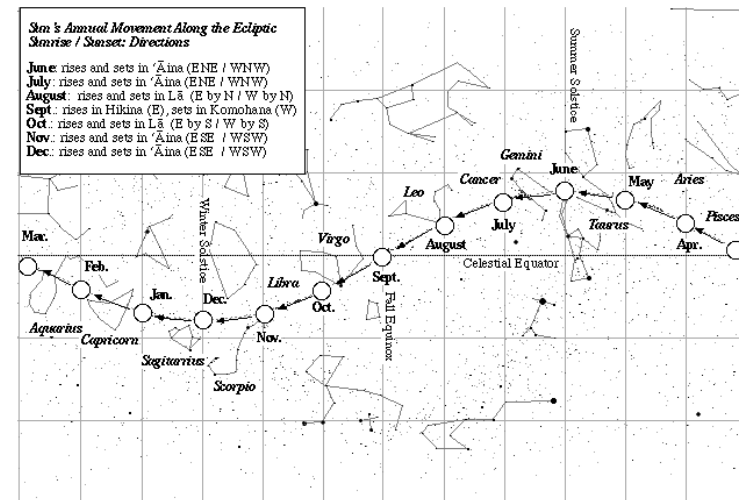
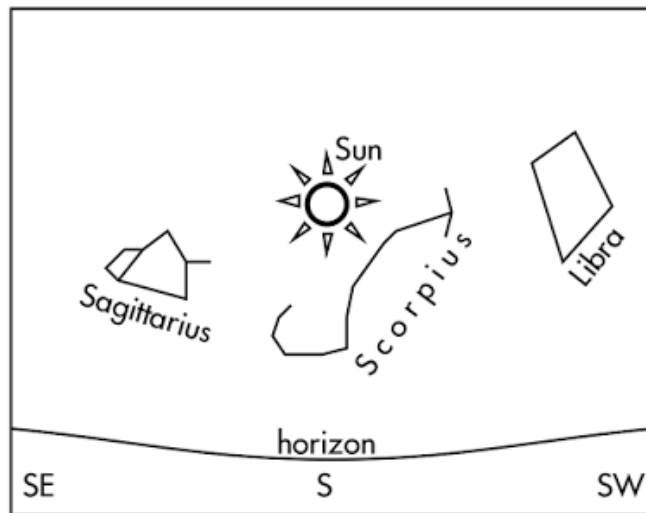
# The Ecliptic



- The Sun's is always found on the **ecliptic**
- The signs of the zodiac are along the ecliptic
- All planets and even the moon found near ecliptic – the plane of the Solar system
- The Earth's axis is tilted 23.5 degrees, so the ecliptic is inclined 23.5 degrees with respect to the celestial equator

# Sun and the Celestial Sphere

- Sun fixed to celestial sphere during one day (almost)
  - The Sun follows the ecliptic throughout the year, not the day!



(Stars and constellations shown here in their correct locations, but they cannot be viewed with the naked eye during the day, as shown)  
 (Animation again)

# QUESTIONS!

LT book, “Ecliptic” starting on Page 13, #1-12

About 10 min

# Yearly motions of the sky

## Goals for Today:

1) How do the star's motions change with latitude?

At poles, all stars are visible. For rest of planet, some stars circumpolar, some rise and set, and some are never visible.

2) What is the significance of the signs of the Zodiac and the ecliptic

They trace the path the Sun makes throughout the year

3) Why do the constellations change throughout the year?

Different parts of the sky are visible at night.